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Throughout, let R be a commutative ring with 1, and M an R -module.

Exercise 1. Let $I \subset R$ be an ideal, M an R -module. Prove $IM = \{\sum a_i m_i : a_i \in I, m_i \in M, \text{ finite sum}\}$ is a submodule of M .

Proof. It is clear that IM is nonempty and closed under addition, as $1 \in IM$ and the sum of two finite sums will also be finite and of the form $\sum a_i m_i$. For any $r \in R$ and $\sum a_i m_i$, we have $r(\sum a_i m_i) = \sum (ra_i) m_i$. Since I is an ideal, $ra_i \in I$. Hence IM is a submodule of M ■

Exercise 2. Suppose M is an R -module, and $X \subset M$ is a subset. Prove that $N = \langle X \rangle$ is the *smallest* submodule containing the set X . First formulate what “smallest” means.

Proof. Suppose Y is a submodule of M such that $X \subset Y \subset N$. Let $\sum rx \in N$. Since $X \subseteq Y$, $x \in Y$. Since Y is a submodule, $\sum rx \in Y$. Thus $N \subseteq Y$; i.e., $Y = N$. ■

Exercise 3. Let V be the 3-dimensional $\mathbb{Q}$ -vector space $\{a_0 + a_1 \sqrt[3]{2} + a_2 \sqrt[3]{4} : a_i \in \mathbb{Q}\}$, and let $T = \sqrt[3]{2} + \sqrt[3]{4}$.

- Convince yourself that left-multiplication by T is a $\mathbb{Q}$ -linear transformation on V .
- Find a matrix for T with respect to the basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$.
- Compute $f(x)v$, where $f(x) = x^2 - 2x - 4$, x acts as the linear transformation T , and $v = 2 + \sqrt[3]{2} + \sqrt[3]{4}$.
- Find a polynomial $p(x)$ such that $p(x)v = 0$ for all v .

Proof. (a) Given $T = \sqrt[3]{2} + \sqrt[3]{4}$ and $v \in V$, write, $v = a_0 + a_1 \sqrt[3]{2} + a_2 \sqrt[3]{4}$, where $a_0, a_1, a_2 \in \mathbb{Q}$. Then:

$$\begin{aligned} T(v) &= (\sqrt[3]{2} + \sqrt[3]{4})(a_0 + a_1 \sqrt[3]{2} + a_2 \sqrt[3]{4}) \\ &= a_0 \sqrt[3]{2} + a_0 \sqrt[3]{4} + a_1 \sqrt[3]{4} + 2a_1 + 2a_2 + 2a_2 \sqrt[3]{2} \\ &= (2a_1 + 2a_2) + (a_0 + 2a_2) \sqrt[3]{2} + (a_0 + a_1) \sqrt[3]{4} \\ &\in V. \end{aligned}$$

Since each coefficient is rational, T takes elements of V to elements of V . Moreover, by associativity and distributivity, for any $v_1, v_2 \in V$ and $q \in \mathbb{Q}$, we have $T(v_1 + qv_2) = Tv_1 + qTv_2$. Therefore, left-multiplication by T is a $\mathbb{Q}$ -linear transformation on V .

(b) Let $\mathcal{B} = (1, \sqrt[3]{2}, \sqrt[3]{4})$. We compute the image of each basis vector as follows:

$$\begin{aligned} T(1) &= 0 \cdot 1 + 1 \cdot \sqrt[3]{2} + 1 \cdot \sqrt[3]{4}, \\ T(\sqrt[3]{2}) &= 2 \cdot 1 + 0 \cdot \sqrt[3]{2} + 1 \cdot \sqrt[3]{4}, \\ T(\sqrt[3]{4}) &= 2 \cdot 1 + 2 \cdot \sqrt[3]{2} + 0 \cdot \sqrt[3]{4}. \end{aligned}$$

Thus:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 2 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 0 \end{pmatrix}.$$

(c) We have:

$$\begin{aligned}
 f(x)v &= (T^2 - 2T - 4)(2 + \sqrt[3]{2} + \sqrt[3]{4}) \\
 &= \left((\sqrt[3]{2} + \sqrt[3]{4})^2 - 2(\sqrt[3]{2} + \sqrt[3]{4}) - 4 \right) (2 + \sqrt[3]{2} + \sqrt[3]{4}) \\
 &= (\sqrt[3]{4} + 4 + 2\sqrt[3]{2} - 2\sqrt[3]{2} - 2\sqrt[3]{4} - 4)(2 + \sqrt[3]{2} + \sqrt[3]{4}) \\
 &= -\sqrt[3]{4}(2 + \sqrt[3]{2} + \sqrt[3]{4}) \\
 &= -2\sqrt[3]{4} - 2 - 2\sqrt[3]{2}.
 \end{aligned}$$

(d) Let $p(x)$ be the characteristic polynomial of $[T]_B$. Then $p(x)v = 0$ for all $v \in V$ by the Cayley–Hamilton theorem. ■

Exercise 4. Suppose $\{M_i : i \in I\}$ is a set of submodules of M

- (a) Prove that $\sum_i M_i$ is a submodule of M , where all sums are finite. Prove that $\sum_i M_i$ is the smallest submodule of M that contains all of the M_i .:w
- (b) Prove that $\bigcap_i M_i$ is a submodule of M .

Proof. Define $N := \sum_i M_i$.

(a) Note $N \neq 0$ since $1 \in N$. Let $\sum m_i, \sum n_i \in N$. The sum of these two elements is clearly a finite sum, hence they are an element of N . Let $r \in R$ and $\sum m_i \in N$. Then $r(\sum m_i) = \sum rm_i$. Since $rm_i \in M_i$ for each i , we have $r(\sum m_i) \in N$. Thus N is a submodule of M . Suppose now that Y is a submodule of M such that $\bigcup_i M_i \subseteq Y \subseteq N$. Let $\sum m_i \in N$. Note that each $m_i \in \bigcup_i M_i \subseteq Y$. Since Y is a submodule, $\sum m_i \in Y$. Hence $Y = N$.

(b) Note $\bigcap_i M_i \neq 0$ because $1 \in M_i$ for all i . Let $m, n \in \bigcap_i M_i$. Then $m, n \in M_i$ for all i . Hence for all $r \in R$, we have $m + rn \in M_i$ for all i . Thus $m + rn \in \bigcap_i M_i$, hence it is a submodule of M . ■

Exercise 5. Suppose $M_1 \subset M_2 \subset M_3 \subset \dots$ is an ascending sequence of submodules of an R -module M . Prove that $\bigcup_i M_i$ is a submodule of M .

Proof. Note that $1 \in \bigcup_i M_i$ because $1 \in M_i$ for all i . If $m, n \in \bigcup_i M_i$, then $m \in M_j$ and $n \in M_k$ for some indices j, k . There exists some index ℓ such that $M_j, M_k \subseteq M_\ell$, hence $m, n \in M_\ell$. Thus $m + n \in M_\ell \subseteq \bigcup_i M_i$. Moreover, for any $r \in R$, we have $mr \in M_\ell \subseteq \bigcup_i M_i$. Thus $\bigcup_i M_i$ is a submodule. ■

Exercise 6. Suppose $f : M \rightarrow M'$ and $g : N \rightarrow N'$ are R -module homomorphisms.

- (a) Prove that f and g induce R -module homomorphisms $f^* : \text{Hom}_R(M', N) \rightarrow \text{Hom}_R(M, N)$ and $g_* : \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M, N')$.
- (b) Prove that if f is onto then f^* is one-to-one, and if g is one-to-one then g_* is one-to-one.

Proof. (a) Define $f^* : \text{Hom}_R(M', N) \rightarrow \text{Hom}_R(M, N)$ by $f^*(\varphi) := f \circ \varphi$. This is indeed an R -module homomorphism, as for any $\varphi_1, \varphi_2 \in \text{Hom}_R(M', N)$ and $r \in R$ we have:

$$\begin{aligned} f^*(\varphi_1 + r\varphi_2)(m) &= ((\varphi_1 + r\varphi_2) \circ f)(m) \\ &= (\varphi_1 + r\varphi_2)(f(m)) \\ &= \varphi_1(f(m)) + r\varphi_2(f(m)) \\ &= ((\varphi_1 \circ f) + r(\varphi_2 \circ f))(m) \\ &= (f^*(\varphi_1) + rf^*(\varphi_2))(m). \end{aligned}$$

Similarly, define $g_* : \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M, N')$ by $g_*(\varphi) := g \circ \varphi$. Then g_* is an R -module homomorphism, as for any $\varphi_1, \varphi_2 \in \text{Hom}_R(M, N)$ and $r \in R$ we have:

$$\begin{aligned} g_*(\varphi_1 + r\varphi_2)(m) &= (g \circ (\varphi_1 + r\varphi_2))(m) \\ &= g((\varphi_1 + r\varphi_2)(m)) \\ &= g(\varphi_1(m) + r\varphi_2(m)) \\ &= (g \circ \varphi_1)(m) + r(g \circ \varphi_2)(m) \\ &= ((g \circ \varphi_1) + r(g \circ \varphi_2))(m). \end{aligned}$$

(b) Suppose $f^*(\varphi_1) = f^*(\varphi_2)$ for some $\varphi_1, \varphi_2 \in \text{Hom}_R(M', N)$. Then $\varphi_1(f(m)) = \varphi_2(f(m))$ for every $m \in M'$. Since f is onto, there exists $x \in M$ such that $f(m) = x$. Thus $\varphi_1(x) = \varphi_2(x)$. Since φ_1 and φ_2 are equal on all elements, we have that $\varphi_1 = \varphi_2$. Thus f^* is one-to-one. Now suppose $g_*(\psi_1) = g_*(\psi_2)$ for some $\psi_1, \psi_2 \in \text{Hom}_R(M, N)$. Then $g(\psi_1(m)) = g(\psi_2(m))$ for all $m \in M$. Since g is one-to-one, $\psi_1(m) = \psi_2(m)$ for all $m \in M$. Since ψ_1 and ψ_2 are equal on all elements, we have $\psi_1 = \psi_2$. Thus g_* is one-to-one. ■

Exercise 7. If N is a submodule of M , the annihilator of N in R is $\text{Ann}_R(N) = \{r \in R : rn = 0 \ \forall n \in N\}$.

(a) Prove $\text{Ann}_R(N)$ is a 2-sided ideal of R .

(b) Suppose $M = \mathbb{Z}_{24} \times \mathbb{Z}_{15} \times \mathbb{Z}_{50}$. Compute $\text{Ann}_{\mathbb{Z}}(M)$.

Proof. (a) Note that $\text{Ann}_R(N) \neq \emptyset$ because $0 \in \text{Ann}_R(N)$. If $r_1, r_2 \in \text{Ann}_R(N)$, then $(r_1 - r_2)n = r_1n - r_2n = 0$, hence $r_1 - r_2 \in \text{Ann}_R(N)$. If $x \in \text{Ann}_R(N)$, and $r \in R$, then $(xr)n = (rx)n = 0$ and $(rx)n = r(xn) = 0$. Thus $xr, rx \in \text{Ann}_R(N)$, implying that it is a 2-sided ideal of R ;

(b) It is enough to determine all $\alpha \in \mathbb{Z}$ such that $\alpha(\overline{1}, \overline{0}, \overline{0}) = 0$, $\alpha(\overline{0}, \overline{1}, \overline{0}) = 0$, and $\alpha(\overline{0}, \overline{0}, \overline{1}) = 0$. This is equivalent to determining all α such that $\overline{\alpha} \equiv 0 \pmod{24}$, $\overline{\alpha} \equiv 0 \pmod{15}$, and $\overline{\alpha} \equiv 0 \pmod{50}$. This system of equations is equivalent to $\overline{\alpha} \equiv 0 \pmod{\text{lcm}(24, 15, 50)}$, thus $\text{Ann}_{\mathbb{Z}}(M) = \langle \text{lcm}(24, 15, 50) \rangle$. ■

Exercise 8. Assume R is commutative. Show that M is an irreducible R -module if and only if M is R -isomorphic to $R/\mathfrak{m}$, for some maximal (2-sided) ideal $\mathfrak{m}$.

Proof. Suppose that M is an irreducible R -module. For any $m \in M$, note that $\langle m \rangle$ is a submodule of M . Hence $M = \langle m \rangle$. Define $\varphi : R \rightarrow M$ by $r \mapsto rm$. This is clearly an R -module homomorphism, as $\varphi(r_1 + r_2) = (r_1 + r_2)m = r_1m + r_2m = \varphi(r_1) + \varphi(r_2)$ and $\varphi(rr_0) = (rr_0)m = r(r_0m) = r\varphi(r_0)$. Treating M and R as groups, by the First Isomorphism Theorem we have $R/\ker \varphi \cong \text{im } \varphi$. Note that φ is surjective, as for any $m \in M = \langle m \rangle$, we have $m = r_0m$ for some $r_0 \in R$; i.e., $\varphi(r_0) = r_0m = m$. Hence $R/\ker \varphi = M$. It remains to show that $\ker \varphi$ is a maximal ideal. Suppose J is an ideal of R such that $\ker \varphi \subseteq J \subseteq R$. By the Third Isomorphism Theorem, we have $(R/\ker \varphi)(J/\ker \varphi) \cong R/J$, and in particular, $J/\ker \varphi$ is an ideal of $R/\ker \varphi \cong M$. Since M is irreducible,

either $J/\ker \varphi = 0$ or $J/\ker \varphi = M$. If $J/\ker \varphi = 0$, then $J = \ker \varphi$. If $J/\ker \varphi = M$, then $J = R$. Thus $\ker \varphi$ is a maximal ideal.

Conversely, let $\varphi : M \rightarrow R/\mathfrak{m}$ be an R -isomorphism for some maximal ideal $\mathfrak{m}$. Let N be a submodule of M . Note that $\varphi(N)$ is an ideal, as it is closed under addition and scalar multiplication by R . Since $\mathfrak{m}$ is maximal, $R/\mathfrak{m}$ is a field. Hence $\varphi(N) = 0$ or $\varphi(N) = R/\mathfrak{m}$, implying that $N = 0$ or $N = M$. Thus M is irreducible. ■

Exercise 9. Suppose $N \subset M$ is a submodule. Prove that if N and M/N are finitely generated, then M is finitely generated.

Proof. If N is finitely generated, then $N = \langle X \rangle$ for some $X \subset N$. Let $m \in M$ be arbitrary and consider $m + \langle X \rangle \in M/\langle X \rangle$. Since $M/\langle X \rangle$ is finitely generated, there is a subset $Y \subset M/\langle X \rangle$ such that $M/\langle X \rangle = \langle Y \rangle$. Hence there exists $r_1, \dots, r_n \in R$ and $y_1 + \langle X \rangle, \dots, y_n + \langle X \rangle \in Y$ such that:

$$\begin{aligned} m + \langle X \rangle &= \sum_{i=1}^n r_i (y_i + \langle X \rangle) \\ &= \left(\sum_{i=1}^n r_i y_i \right) + \langle X \rangle. \end{aligned}$$

This means $m - \left(\sum_{i=1}^n r_i y_i \right) \in \langle X \rangle$, so there exists $s_1, \dots, s_k \in R$ and $x_1, \dots, x_k \in X$ such that:

$$m - \left(\sum_{i=1}^n r_i y_i \right) = \sum_{i=1}^k s_i x_i.$$

Thus $m = \left(\sum_{i=1}^n r_i y_i \right) + \left(\sum_{i=1}^k s_i x_i \right)$. Since $m \in M$ was arbitrary, M is finitely generated. ■